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Optimizing the controller placement problem in SDN with uncertain parameters with robust optimization

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***Abstract*— By separating the network management plane from the data plane, Software-Defined Networking (SDN) is a networking architecture that offers a centralized view of the network. But due to the growth of the network, there must be several controllers. The number of them and their location create the issue of the controller placement problem (CPP). In many studies conducted in this field, it has been assumed that network information is fully available. In practice, due to dynamic conditions, the obtained information has uncertainty. In facing this challenge of uncertainty, there are different approaches, but one effective approach has been robust optimization techniques. In robust optimization, the goal is to make a feasible and optimal decision to optimize the objective function in the worst case. Robust methods are also different, and we use the Bertsimas method. In this article, the uncertainty of the traffic rate parameters produced by each switch as well as the capacity or processing power parameter of each controller will be the problem. Finally, we will show the advantage of the robust method in a scenario. In such a way that if there is a prediction for the deviations that happen during the execution of the network, the actual value of the objective function will be equal or even reduced in some cases.**

Keywords*— *software-Defined Networking (SDN); controller placement problem (CPP); robust optimization

I. INTRODUCTION

The Internet of Things (IoT) has led to the development of a new generation of networks that have connected a large number of devices and created a new level of interaction and responsiveness [1]. New computer networks consist of many devices such as routers, gateways, firewalls, proxies, and even load balancers, each of which is made by different companies. As a result, they may run on special software, which causes complexity in network management. Software-based networking has ushered in a new era in the networking industry. The goal of software-defined networking is to provide a programmable network that is managed by software. By separating the control plane from the data plane, SDN simplifies networking and offers new possibilities. The SDN controller, as a centralized entity, manages the entire network based on the policies desired by the administrator. At the higher level of the controller, there are intelligent programs that form the program level [2]. Software-Defined Networking has gained popularity as a new network architecture in recent years. However, the growing size of networks has created challenges for the centralized control plane approach. As a result, distributed control plane has been considered as a solution to meet network needs. To improve the performance of distributed SDN, optimization algorithms are applied to address the controller placement problem. This application of optimization algorithms has the potential to improve the overall efficiency and effectiveness of distributed control plane [3]. The controller placement problem is an important challenge in SDN. The deployment of a properly designed SDN network is required to identify the optimal number of controllers to increase network performance. However, the best possible controller placement to increase network performance faces many challenges [4]. The CPP can be examined through a number of objectives, including load balancing, fault tolerance, resilience, and network responsiveness. It frequently entails weighing a number of competing goals and trade-offs. The literature has been interested in modeling controller placement problems using graph theory, game theory, and queueing theory. The controller placement problem has a significant impact on a number of issues in network design. The network design goal, which affects latency, resilience, and QoS, determines where to locate the controllers for effective control [5].

Most of the articles written about the controller placement problem have considered the parameters of that problem to be fixed and deterministic. For example, they assume static network traffic while it is dynamically affected by time, location, economic levels, and regional development. Also, the capacity or processing power of a controller is not fixed and certain in practice due to the variability of the usage of the controller's resources such as the CPU, memory, etc. The irreversibility of controller placement results in high costs for repositioning. Ignoring the uncertainty in the problem parameters can lead to theoretically suboptimal solutions, but in real-life scenarios, these solutions cannot handle sudden changes, leading to wasted resources or controller failure [6]. One of the ways to deal with uncertainty in optimization problems is through robust optimization. In this article, we solve and analyze the CPP problem by considering uncertain parameters with the help of robust optimization.

In the rest of the article, in Section 2, we will review the relevant work done in the past. In Section 3, we will examine robust optimization, and in Section 4, we will formulate the CPP

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problem with the help of robust optimization. In Section 5, the formulated problem is analyzed with some examples. In Section 6, the advantage of using robust optimization during a scenario is examined. Finally, the conclusion of the article will come.

II. Related work

In [7], the idea of the location of the dynamic controller and the dynamic assignment of the switch to the controller in order to increase effectiveness and manage traffic load changes is considered. The traffic-aware controller placement (TACP) problem, expressed as an integer linear program (ILP), involves optimizing load balancing between controllers. The Dynamic Placement Fastest Assignment (DPFA) method, which uses a counting-based ideal algorithm, is presented as a solution to this issue. Huque [8] proposed a hybrid solution to the controller placement problem and the controller provisioning problem by considering the location of the switches and setting the delay constraint that occurs due to dynamic traffic. Their solution was circular partitioning of the data plane, which divides the network area into a number of sub-areas and implements the smallest enclosing disk algorithm for each set of switches located in each sub-area. Dixit [9] proposed a method to dynamically distribute switch traffic to several controllers and reduce the overload imposed on the controller. They did this by designing a controller capable of detecting dynamic traffic. Tanha [10] also considered traffic dynamically in her article. In this article, at the same time that each switch is assigned a main controller, it also assigns one or more backup controllers. When the main controller of a switch is overloaded and fails, the backup controller takes over the responsibility of the switch. The author's solution is to use a clique-based approach in graph theory. Minimizing the operation cost of the control plane and minimizing its setup cost were the main concerns of the authors in [11] about the control plane in dynamic traffic scenarios. In order to solve this problem, they developed a multi-period offline optimization model with the aim of reducing the overall cost associated with implementing flow regulation and control level adaptation. In the paper [12], a robust controller placement model considering dynamic traffic is proposed to reduce switch migration. This model is transformed into a linear programming problem using strong duality theory and the tabu search algorithm. The simulation results show that different placement strategies can be obtained.

In this article, we consider the capacity of each controller as non-deterministic at the same time as the traffic is non-deterministic, and with the help of robust optimization, we will formulate the problem and then analyze it.

III. Robust optimization

Real-world optimization problems often have uncertain data that is often unknown at the time of solution. This uncertainty comes from measurement and estimation errors due to the impossibility of accurate measurement or estimation of physical systems, technological processes, or environmental conditions. Real-world optimization applications face the possibility that uncertainty in even small data sets can render the nominally optimal solution practically meaningless. Therefore, optimization requires a method to detect cases where data uncertainty significantly affects the quality of the nominal solution and to create reliable solutions against this uncertainty [13]. To model uncertainty, which is a set function satisfying specific requirements, one needs the mathematical tool of uncertainty. In 2007, Liu established a theory of uncertainty based on the axioms of self-duality, countable subadditivity, normalcy, monotonicity, and product measure. For hazy phenomena, this area of mathematics offers a mathematical model [14].

To develop models that are immune to data uncertainty, in the first step, Soyster presented a method that proposes a linear optimization model for feasible solutions in convex sets. However, this model produces conservative solutions and sacrifices robustness for optimization. In this way, all the input data is considered uncertain, and as a result, the answer is far from the nominal optimal answer [15]. Next, Ben-Tal and Nemirovski developed a robust optimization theory considering uncertain linear problems where these uncertain data belong to an elliptic set. This method can estimate more complex uncertainty sets, but results in non-linear (but convex) models that are computationally more complex than Soyster's previous linear model [13].

But Bertsimas and Sim's method provides a newer approach to robust linear optimization while retaining the advantages of Soyster's linear framework. This method provides full control over conservatism for each constraint and protects against constraint violations deterministically when only a few input data points are uncertain. This method guarantees that a robust solution is possible with high probability. Also, even if other uncertain data is available beyond the number of assumed uncertain input data, this proposed framework is at least as flexible as the Ben-Tal and Nemirovski method and can also be used for discrete optimization problems [16].

Considering the advantages of the Bertsimas and Sim method, which include maintaining the linear framework of the problem, complete control over conservatism, and the ability to adjust the number of uncertain inputs of the problem (or the gamma parameter) by the network operator, in the CPP problem with uncertain parameters, this method will continue to be used.

IV. Robust optimization of the CPP

In this part, we will consider the problem of positioning the controller with uncertain parameters, and according to the previous part, we will formulate the problem with the help of robust optimization. In this problem, the data layer is divided into a number of domains, and there are a number of switches in each domain. The management of each domain is done by the controller that has the capacity to manage that domain according to the number of switches in it. In this problem, there are two variables. The first variable, x, shows whether a specific controller is placed in a specific position or domain (a value of one) or not (a value of zero). and the second variable y, which determines which position each switch will be placed in according to the capacity of the controllers and is managed by the controller of that position. (If it is placed in that position, it will have a value of one, and otherwise it will have a value of zero). The parameters of this problem are r, which indicates the traffic rate produced by each switch, and μ, which indicates the capacity or processing rate of each controller. Our intention in

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this article is to make these two parameters uncertain. As mentioned earlier, the traffic rate is dynamically affected by time, place, economic levels, and regional development, and the capacity or processing power of a controller is also affected by the use of its resources. Also, there are two parameters ω and ρ that show the cost of installing each controller and the cost of power per bit, respectively, and of course these two parameters are considered fixed due to the fact that they are determined by the operator. Symptoms are listed in Table 1.

TABLE1 SYMPTOMS

Problem parameter	symbols
Number of switches	n
Number of controllers	m
The traffic rate	r
capacity of the controller	μ
The cost of installing the controller	ω
Power cost per bit	ρ
Position binary variable	x
Switch binary variable	y
Variables of dual problem	a,B
the degree of conservatism	Γ

The purpose of this issue in equation (1) is to reduce the placement cost and guarantee the delay between a controller and a switch. The limitations of this problem are as follows: only one controller is placed in each position, and each switch must be under the control and management of only one controller. Also, the total number of requests assigned to a controller should not exceed the processing capacity of that controller. As mentioned, the two variables will only take values of one and zero and According to the explanation, the problem is formulated as follows:

$$\begin{aligned} \min\ & \omega \textstyle\sum_p x_{cp} + \rho \sum_s y_{sp} r_{sp} \\ \text{s.t.}\ & \textstyle\sum_c x_{cp} \le 1\ ,\forall_p \\ & \textstyle\sum_p y_{sp} = 1\ ,\forall_s \\ & \textstyle\sum_s y_{sp}\, r_{sp}\ \mu\, x_{cp}\ ,\forall_p \\ & x_{cp} \in \{0,1\}\ ,\forall_c\ \forall_p \\ & y_{sp} \in \{0,1\}\ ,\forall_s\ \forall_p \end{aligned} \tag{1}$$

In the first step, we will consider the r parameter uncertain and use the robust optimization solution (Bertsimas and Sim method). If we denote the parameter of the production traffic rate considered for each switch with r_{sp} and the actual value of the traffic rate during the execution of the problem with $\tilde{r}_{sp}$, the value of the production traffic rate considered will have a deviation, which we denote by $\hat{r}_{sp}$. As a result, we will have: $r_{sp} - \hat{r}_{sp} \le \tilde{r}_{sp} \le r_{sp} + \hat{r}_{sp}$.

Therefore, in the above equation, we must put the actual value of the traffic rate in the objective function (that is, $\omega \sum_p x_{cp} + \rho \sum_p y_{sp} \tilde{r}_{sp}$. From this real value, there are values that are certain, and there are also values that we are not sure about. Therefore, instead of the objective function, we will have: $\omega \sum_p x_{cp} + \rho(\sum_s y_{sp} \tilde{r}_{sp} + \sum_s \eta_{sp} \hat{r}_{sp} y_{sp})$. The η_{sp} variable is also for normalization and is defined as follows: $\eta_{sp} = \frac{(\tilde{r}_{sp} - r_{sp})}{\hat{r}_{sp}}$.

In the method of Bertsimas and Sim, for the uncertain part, for each switch, it forms all the possible subsets (l_p) of the index of positions (J_p), but only selects the subsets whose number is equal to gamma (Γ) and calculates their values. And among them, it returns the largest value and adds it to the certainly value. Therefore, the initial equation, taking into account the uncertainty in the production traffic rate parameter, is as follows:

$$\begin{aligned} \min\ & \omega \textstyle\sum_p x_{cp} + \rho(\sum_s y_{sp} r_{sp} + \max_{\{l_p \subseteq J_p, |l_p| = \Gamma_p\}} \{\sum_{s \epsilon l_p} \eta_{sp} \hat{r}_{sp} y_{sp}\}) \\ \text{s.t.}\ & \textstyle\sum_c x_{cp} \le 1\ ,\forall_p \\ & \textstyle\sum_p y_{sp} = 1\ ,\forall_s \\ & \textstyle\sum_s y_{sp}\, r_{sp} \le \mu\, x_{cp}\ ,\forall_p \\ & x_{cp} \in \{0,1\}\ ,\forall_c\ \forall_p \\ & y_{sp} \in \{0,1\}\ ,\forall_s\ \forall_p \end{aligned} \tag{2}$$

Considering that the existence of maximization in the equation is not possible and causes the program to be non-convex, the expression $\max_{\{l_p \subseteq J_p, |l_p| = \Gamma_p\}} \{\sum_{s \epsilon l_p} \eta_{sp} \hat{r}_{sp} y_{sp}\}$ is equivalent to another program named $\beta_p = (y^*_{sp}, \Gamma_p)$ and then we put its dual in the equation. We have:

$$\begin{aligned} \beta_p = (y^*_{sp}, \Gamma_p) = \max\ & \textstyle\sum_{s \epsilon J_p} \hat{r}_{sp} y^*_{sp} Q_{sp} \\ \text{s.t.}\ & \textstyle\sum_{s \epsilon J_p} Q_{sp} \le \Gamma_p \\ & 0 \le Q_{sp} \le 1\ ,\ \forall_{s \epsilon J_p} \end{aligned} \tag{3}$$

In this way, assuming that the r_{sp} matrix has the positions within the possible subsets (l_p) of the position index (J_p) (according to the gamma value), it is possible to store them or vice versa, which is done by the Q_{sp} variable. The dual of the above program (a and B are dual variables) is equal to:

$$\begin{aligned} \min\ & \textstyle\sum_{s \epsilon J_p} B_{sp} + \Gamma_p a_p \\ \text{s.t.}\ & a_p + p_{sp} \ge \hat{r}_{sp} y_{sp}\ ,\forall_{s,p \epsilon J_p} \\ & B_{sp} \ge 0\ ,\forall_{s \epsilon J_p} \\ & a_p \ge 0\ ,\forall_p \end{aligned} \tag{4}$$

Finally, the robust CPP problem with the uncertain parameter of the generated traffic rate will be as follows (given that the traffic rate parameter is also included in the fourth condition of the initial problem, all these steps will be repeated for that condition as well):

$$\begin{aligned} \min\ & \omega \textstyle\sum_p x_{cp} + \rho \sum_s y_{sp} r_{sp} + \sum_{s \epsilon J_p} B_{sp} + \Gamma_p a_p \\ \text{s.t}\ & \textstyle\sum_c x_{cp} \le 1 \\ & \textstyle\sum_p y_{sp} = 1 \\ & \textstyle\sum_s y_{sp}\, r_{sp} + \sum_{s \epsilon J_p} B2_{sp} + \Gamma_p a2_p \le \mu\, x_{cp} \end{aligned}$$

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$$
\begin{aligned}
& x_{cp} \epsilon \{0,1\} \ , \forall_c \ \forall_p \\
& y_{sp} \epsilon \{0,1\} \ , \forall_s \ \forall_p \\
& a_p + B_{sp} \geq \hat{r}_{sp} y_{sp} \ , \forall_{p,s\epsilon J_p} \\
& B_{sp} \geq 0 \ , \forall_{s\epsilon J_p}, \forall_p \qquad (5) \\
& a_p \geq 0 \ , \forall_p \\
& a2_p + B2_{sp} \geq \hat{r}_{sp} y2_{sp} \ , \forall_{p,s\epsilon J_p} \\
& B2_{sp} \geq 0 \ , \forall_{s\epsilon J_p}, \forall_p \\
& a2_p \geq 0 \ , \forall_p
\end{aligned}
$$

In the same way, the robust problem with the uncertain parameter of the controller capacity will be as follows:

$$
\begin{aligned}
min \ \ & \omega \textstyle\sum_p x_{cp} + \rho \sum_s y_{sp} r_{sp} \\
\text{s.t.} \ \ & \textstyle\sum_c x_{cp} \leq 1 \ , \forall_p \\
& \textstyle\sum_p y_{sp} = 1 \ , \forall_s \\
& \textstyle\sum_s y_{sp} r_{sp} \leq \mu_{cp} x_{cp} + \sum_{c\epsilon J_p} B_{cp} + \Gamma_p a_p \ , \forall_p \\
& x_{cp} \epsilon \{0,1\} \ , \forall_c \ \forall_p \qquad (6) \\
& y_{sp} \epsilon \{0,1\} \ , \forall_s \ \forall_p \\
& a_p + B_{sp} \geq \hat{\mu}_{cp} x_{cp} \ , \forall_{p,s\epsilon J_p} \\
& B_{sp} \geq 0 \ , \forall_{s\epsilon J_p}, \forall_p \\
& a_p \geq 0 \ , \forall_p
\end{aligned}
$$

The combination of these two problems will also show the CPP problem with uncertain parameters.

$$
\begin{aligned}
min \ \ & \omega \textstyle\sum_p x_{cp} + \rho \sum_s y_{sp} r_{sp} + \sum_{s\epsilon J_p} B1_{sp} + \Gamma_p a1_p \\
\text{s.t} \ \ & \textstyle\sum_c x_{cp} \leq 1 \\
& \textstyle\sum_p y_{sp} = 1 \\
& \textstyle\sum_s y_{sp} r_{sp} + \sum_{s\epsilon J_p} B2_{sp} + \Gamma_p a2_p \leq \mu_{cp} x_{cp} + \sum_{c\epsilon J_p} B3_{cp} + \Gamma_p a3_p \\
& x_{cp} \epsilon \{0,1\} \ , \forall_c \ \forall_p \\
& y_{sp} \epsilon \{0,1\} \ , \forall_s \ \forall_p \qquad (7) \\
& a1_p + B1_{sp} \geq \hat{r}_{sp} y_{sp} \ , \forall_{p,s\epsilon J_p} \\
& B1_{sp} \geq 0 \ , \forall_{s\epsilon J_p}, \forall_p \\
& a1_p \geq 0 \ , \forall_p \\
& a2_p + B2_{sp} \geq \hat{r}_{sp} y2_{sp} \ , \forall_{p,s\epsilon J_p} \\
& B2_{sp} \geq 0 \ , \forall_{s\epsilon J_p}, \forall_p \\
& a2_p \geq 0 \ , \forall_p \\
& a3_p + B3_{sp} \geq \hat{\mu}_{cp} x_{cp} \ , \forall_{p,s\epsilon J_p} \\
& B3_{sp} \geq 0 \ , \forall_{s\epsilon J_p}, \forall_p \\
& a3_p \geq 0 \ , \forall_p
\end{aligned}
$$

V. Robust Optimization Analysis of CPP

In this part, with two examples, we will examine the problem of optimized resistant CPP and analyze the change in gamma value or the degree of conservatism. To better clarify this analysis, the problem space is considered small, and the values given to the parameters are considered as units instead of the actual values in the network world to help further understanding. Table 2 shows the values given to the parameters.

TABLE 2 VALUES GIVEN

parameter	given values
Number of controllers (c)	3
Number of positions (p)	3
Number of switches (s)	5
Gamma value (Γ)	1,2,3,4
traffic rate (r_s)	100 units
Traffic rate deviation ($\hat{r}_s$) amount	10 units
Capacity deviation ($\hat{\mu}_c$) amount	10 units
Capacity of each controller (μ)	respectively [100;200;200] units

First example: it is assumed that the number of three controllers should be placed in three positions and manage the number of five switches. After the implementation of the initial problem, the variables are specified, and the location of the controllers and the location of the switches in the positions to be managed are determined. Fig. 1 shows the final result of this problem. Also, after the implementation of the initial problem, the value of the objective function will be equal to 503 (five switches with 100 traffic units plus the total number of controllers, which is equal to 3, of course. With the costs that will be applied by the network manager, this value of the objective function will be normalized).

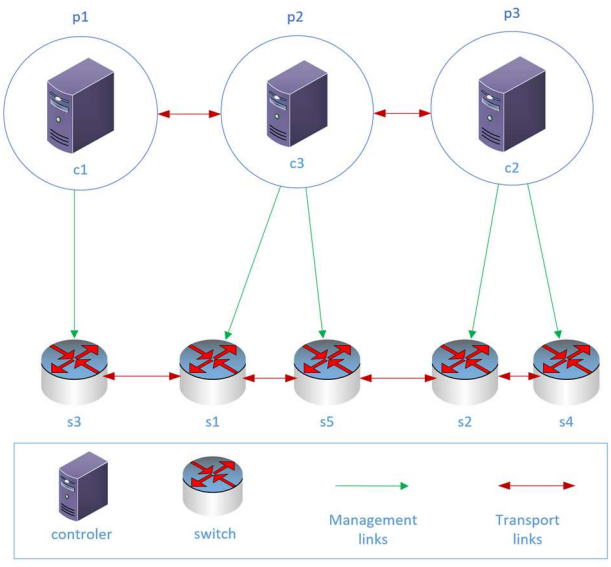

FIGURE 1 FINAL RESULT OF FIRST EXAMPLE AFTER IMPLEMENTATION

a) The uncertain parameter of traffic rate: Now, to analyze the robust optimization problem for the uncertain parameter of traffic rate, it is initially assumed that the gamma value is equal to zero, which means that no deviation is considered for the

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production rate of the switches. According to this situation, the value of the objective function will be equal to 503 because there is no change in the problem. By setting the gamma value equal to one, it will mean that at most one switch will be biased in each position. Considering that there is at least one switch in each position, therefore there are 10 units of deviation in each position and as a result, 30 units will be added to the value of the objective function (533). Now, if the gamma value is added by one more unit, that is, at most 2 switches have a deviation in each position, the value of 60 units should be added to the objective function, but since the total number of switches is five and there is only one switch in the first position, finally the value of the objective function will be increased by 50 units (553). From now on, adding the value of gamma (3, 4, etc.) will not change the objective function, because with gamma 2, all switches are considered to have deviation. Table 3 shows the output values.

TABLE 3 OUTPUT VALUES ACCORDING TO THE GAMMA VALUE

Gamma value	0	1	2	3	4
The value of the objective function	503	533	553	553	553

b) The uncertain parameter of the controller's capacity: Now we examine the robust problem with the uncertain parameter of the controllers' capacity in this example. With the value of gamma equal to zero, according to the previous explanation, there is no change in the problem, and the value of the objective function will remain the same as 503. With a value of gamma equal to one, it means that in every position, the capacity of a controller will have a deviation. Considering that there is a minimum and maximum controller in each position, finally the value of 30 units will be added to the value of the objective function (533), and from now on the addition of the gamma value will not change the value of the objective function. Table 4 shows the output values.

TABLE 4 OUTPUT VALUES ACCORDING TO THE GAMMA VALUE

Gamma value	0	1	2	3
The value of the objective function	503	533	533	533

Finally, as it was determined, the maximum amount of deviation that will be added to the value of the objective function will be equal to the same number of switches or controllers that will have deviation.

Second example: In this example, we add the number of switches and put the number 9. The controllers will be the same three numbers, but the capacity of the controllers must be increased to manage these nine switches (200, 300, and 400, respectively). Fig. 2 shows the final result of this problem.

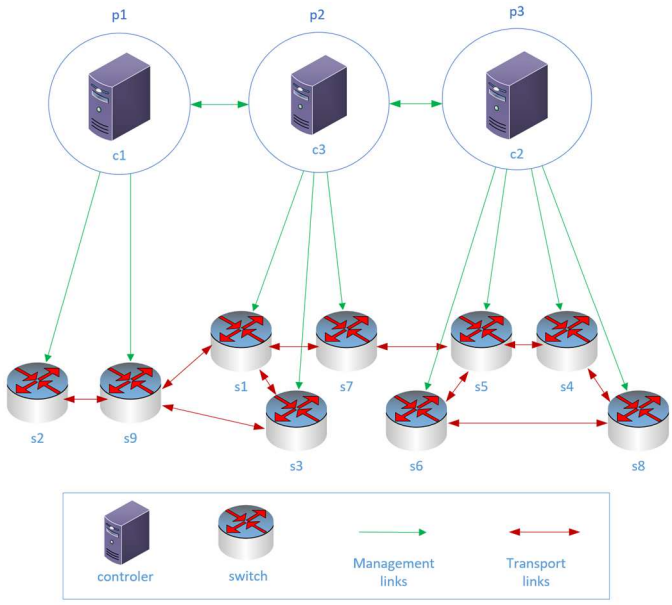

FIGURE 2 FINAL RESULT OF SECOND EXAMPLE AFTER IMPLEMENTATION

By increasing the number of switches, the value of the objective function will naturally increase (903). A gamma value of zero will not change the problem. A gamma value of one means that in each position a switch has a deviation, and 30 units will be added to the value of the objective function (933). With a gamma of 2, 60 units will be added (963). Gamma 3 also means that three switches have deviation in each position, but considering that the first position has only two switches, the value of deviation will be equal to 80 (objective function value 983), and finally, with gamma 4, all switches with deviation will be considered (993). Table 5 shows the output values.

TABLE 5 OUTPUT VALUES ACCORDING TO THE GAMMA VALUE

Gamma value	0	1	2	3	4	5
The value of the objective function	903	933	963	983	993	993

Therefore, it should be noted that increasing the value of gamma will only affect the items that are in that position. Considering that the number of controllers is three, the analysis of this example for the non-deterministic parameter of controller capacity is the same as the previous example.

VI. INCREASE EFFICIENCY WITH ROBUST OPTIMIZATION

According to the analysis done in the previous part, considering the value of deviation for the problem increases the value of the objective function. But the purpose of solving the robust program is to assume that the estimated value may have an error. Therefore, the intention is that if this value is wrong, it should be prepared for that situation, the deviation should be considered in advance, and a solution resistant to the deviation will be obtained. The objective function is a criterion for evaluating answers. Now it is possible to know how much the total cost can be minimized with the obtained answers. Due to the fact that the real value is not known, it is not possible to get an exact answer. The only value that is known at the moment is

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its estimated value, and therefore the problem can be solved for the estimated value. During network implementation, the actual value of the generated traffic rate is determined. Therefore, you can understand how suitable the answers that have been used are. If the estimated value is correct (that is, the actual production traffic rate of each switch is equal to the estimated rate), the objective function value of the resilient problem will be greater than the objective function value of the original problem. In cases where the actual traffic rate deviates and this deviation is considered in advance, the value of the resistant objective function becomes less or equal to the objective function of the main program.

Below is an example that shows these explanations. In Fig. 3, three controllers control five switches in each stage (each stage assumes higher traffic rates for each switch than the previous stage, and the test is repeated). If, at this stage, the objective function of the original method is compared with the objective function of the robust method, the objective function of the robust method will be lower. But during the execution of the test, the real values of the production traffic rate are determined, and now if we do the test again and this time put the real value of the traffic rate in the test, the objective function of the robust method will be equal to or less than the objective function of the original method. Therefore, according to the above explanations, if we predict the deviation, the objective function of the robust method will be equal or even less than the non- robust method.

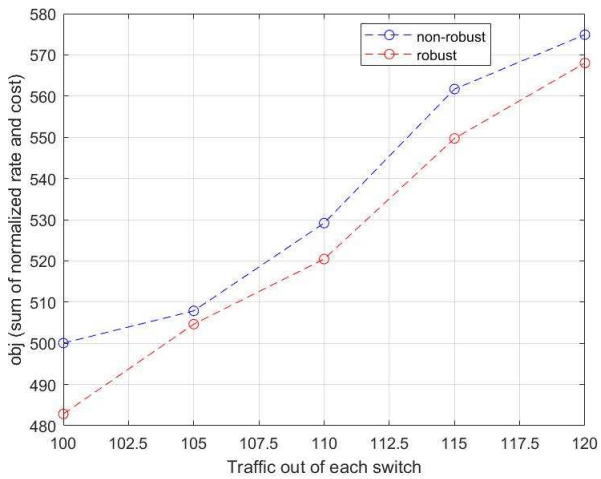

FIGURE 3 BENEFIT OF ROBUST OPTIMIZTION

VII. CONCLUSION

In this article, we considered the controller placement problem with the uncertainty rate of production traffic of each switch and the uncertainty capacity of each controller and formulated this problem with the help of robust optimization techniques. In this article, the types of robust optimization were examined, and we chose the Bertsimas method due to its advantages over other cases. Also, in the following, we will analyze these problems formulated by the robust method with some examples. In these examples, by increasing conservatism in uncertainty parameters, we will examine the change in the value of the objective function. Finally, we showed the advantage of using robust optimization with a scenario.

REFERENCES

[1] H. Adhami, M. Alja'afreh, M. Hoda, J. Zhao, Y. Zhou, and A. El Saddik, "Suitability of SDN and MEC to facilitate digital twin communication over LTE-A," Digit. Commun. Networks, 2023.

[2] A. Liatifis, P. Sarigiannidis, V. Argyriou, and T. Lagkas, "Advancing SDN from OpenFlow to P4: A Survey," ACM Comput. Surv., vol. 55, no. 9, Jan. 2023.

[3] J. Ma, J. Chen, L. Dong, and X. Jiang, "Research on placement of distributed SDN multiple controllers based on IAVOA," Cluster Comput., 2023.

[4] J. Hemagowri and P. Tamil Selvan, "A Hybrid Evolutionary Algorithm of Optimized Controller Placement in SDN Environment," Comput. Assist. Methods Eng. Sci. Online FirstDO - 10.24423/cames.498 , May 2023.

[5] B. B. Gupta and M. Quamara, "An overview of Internet of Things (IoT): Architectural aspects, challenges, and protocols," Concurr. Comput. Pract. Exp., vol. 32, no. 21, p. e4946, 2020.

[6] Z. Zhang, J. Lu, and H. Chen, "Controller robust placement with dynamic traffic in software-defined networking," Comput. Commun., vol. 194, pp. 458–467, 2022.

[7] D. K. Luong, Y.-F. Hu, J.-P. Li, F. Benamrane, M. Ali, and K. Abdo, "Traffic-aware Dynamic Controller Placement using AI techniques in SDN-based aeronautical networks," in 2019 IEEE/AIAA 38th Digital Avionics Systems Conference (DASC), pp. 1–8, 2019.

[8] M. T. I. ul Huque, W. Si, G. Jourjon, and V. Gramoli, "Large-Scale Dynamic Controller Placement," IEEE Trans. Netw. Serv. Manag., vol. 14, no. 1, pp. 63–76, 2017.

[9] A. Dixit, F. Hao, S. Mukherjee, T. V Lakshman, and R. Kompella, "Towards an Elastic Distributed SDN Controller," SIGCOMM Comput. Commun. Rev., vol. 43, no. 4, pp. 7–12, Aug. 2013.

[10] M. Tanha, D. Sajjadi, R. Ruby, and J. Pan, "Capacity-Aware and Delay-Guaranteed Resilient Controller Placement for Software-Defined WANs," IEEE Trans. Netw. Serv. Manag., vol. 15, no. 3, pp. 991–1005, 2018.

[11] M. He, A. Varasteh, and W. Kellerer, "Toward a Flexible Design of SDN Dynamic Control Plane: An Online Optimization Approach," IEEE Trans. Netw. Serv. Manag., vol. 16, no. 4, pp. 1694–1708, 2019.

[12] Z. Zhang, J. Lu, and H. Chen, "Controller robust placement with dynamic traffic in software-defined networking," Comput. Commun., vol. 194, pp. 458–467, 2022.

[13] A. Ben-Tal, L. El Ghaoui, and A. Nemirovski, Robust optimization, vol. 28. Princeton university press, 2009.

[14] B. Liu, "Uncertainty Theory," in Uncertainty Theory: A Branch of Mathematics for Modeling Human Uncertainty, Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 1–79, 2010.

[15] Soyster, A.L. "Convex programming with set-inclusive constraints and applications to inex-act linear programming," Oper. Res., vol.21, pp. 1154-1157, 1973.

[16] D. Bertsimas and M. Sim, "The price of robustness," Oper. Res., vol. 52, no. 1, pp. 35–53, 2004.

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